Product, Quotient, and Trigonometric Derivatives

ANSWER KEY

Section 1: The Product Rule

1 Find the derivative of $f(x) = x^3 \sin x$.

a. $3x^2 \sin x + x^3 \cos x$ ✓

b. $3x^2 \cos x$

c. $3x^2 \sin x - x^3 \cos x$

d. $x^2(3\sin x + \cos x)$

Product rule: $\frac{d}{dx}[uv] = u'v + uv'$. Here $3x^2 \sin x + x^3 \cos x$.

2 If $h(x) = f(x)g(x)$ with $f(2) = 3$, $f'(2) = -1$, $g(2) = 4$, and $g'(2) = 5$, what is $h'(2)$?

a. 11 ✓

b. 19

c. -14

d. 15

$h'(2) = f'(2)g(2) + f(2)g'(2) = (-1)(4) + (3)(5) = -4 + 15 = 11$.

3 Find $\frac{d}{dx}[e^x \cos x]$.

a. $e^x(\cos x - \sin x)$ ✓

b. $e^x(\cos x + \sin x)$

c. $-e^x \sin x$

d. $e^x \cos x$

Product rule: $e^x \cos x + e^x(-\sin x) = e^x(\cos x - \sin x)$.

Section 2: The Quotient Rule

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4 Differentiate $y = \frac{2x-3}{x+4}$.

a. $\frac{11}{(x+4)^2}$ ✓

b. $\frac{5}{(x+4)^2}$

c. $\frac{2x+5}{(x+4)^2}$

d. $\frac{2}{(x+4)^2}$

Quotient rule: $\frac{2(x+4)-(2x-3)(1)}{(x+4)^2} = \frac{2x+8-2x+3}{(x+4)^2} = \frac{11}{(x+4)^2}$.

5 Let $k(x) = \frac{f(x)}{g(x)}$. If $f(1) = 6, f'(1) = 2, g(1) = 3,$ and $g'(1) = -1$, find $k'(1)$.

a. $\frac{4}{3}$ ✓

b. 0

c. 2

d. $-\frac{4}{3}$

$k'(1) = \frac{f'(1)g(1)-f(1)g'(1)}{[g(1)]^2} = \frac{2(3)-6(-1)}{3^2} = \frac{6+6}{9} = \frac{12}{9} = \frac{4}{3}$.

6 Find the derivative of $f(x) = \frac{e^x}{x^2}$.

a. $\frac{e^x(x-2)}{x^3}$ ✓

b. $\frac{e^x(x+2)}{x^3}$

c. $\frac{e^x}{2x}$

d. $\frac{e^x(x-1)}{x^3}$

Quotient rule: $\frac{e^x(x^2)-e^x(2x)}{x^4} = \frac{xe^x(x-2)}{x^4} = \frac{e^x(x-2)}{x^3}$.

Section 3: Derivatives of Basic Trigonometric Functions

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7 What is $\frac{d}{dx}[\tan x]$?

a. $\sec^2 x$ ✓

b. $\sec x \tan x$

c. $-\csc^2 x$

d. $\cot x$

By definition, $\frac{d}{dx}[\frac{\sin x}{\cos x}] = \frac{\cos^2 x + \sin^2 x}{\cos^2 x} = \frac{1}{\cos^2 x} = \sec^2 x$.

8 What is $\frac{d}{dx}[\sec x]$?

a. $\sec x \tan x$ ✓

b. $\sec^2 x$

c. $-\csc x \cot x$

d. $\tan^2 x$

By the quotient rule on $\frac{1}{\cos x}$: $\frac{0 - (-\sin x)}{\cos^2 x} = \frac{\sin x}{\cos^2 x} = \sec x \tan x$.

9 What is $\frac{d}{dx}[\cot x]$?

a. $-\csc^2 x$ ✓

b. $\csc^2 x$

c. $-\sec^2 x$

d. $-\csc x \cot x$

$\frac{d}{dx}[\cot x] = -\csc^2 x$.

Section 4: Higher-Order Derivatives & Acceleration

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10 Find the second derivative $f''(x)$ of $f(x) = \sin x$.

a. $-\sin x$ ✓

b. $\sin x$

c. $-\cos x$

d. $\cos x$

$f'(x) = \cos x$. Then $f''(x) = \frac{d}{dx}[\cos x] = -\sin x$.

11 If position is $s(t) = t^4 - 2t^3 + 5$, what is the acceleration $a(t)$ at $t = 2$?

a. 24 ✓

b. 36

c. 12

d. 48

Velocity $v(t) = 4t^3 - 6t^2$. Acceleration $a(t) = 12t^2 - 12t$. At $t = 2$:
$a(2) = 12(4) - 12(2) = 48 - 24 = 24$.

12 What is the 50th derivative of $f(x) = \cos x$?

a. $-\cos x$ ✓

b. $\cos x$

c. $\sin x$

d. $-\sin x$

Derivatives of cosine repeat in cycles of 4: $f^{(0)} = \cos x$, $f^{(1)} = -\sin x$, $f^{(2)} = -\cos x$, $f^{(3)} = \sin x$. Since $50 = 4(12) + 2$, $f^{(50)}(x) = f^{(2)}(x) = -\cos x$.

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